Task 1: Threat Hunting Practice

Objective

Proactive threat hunting using hypothesis-driven methodology to detect potential abuse of privileged account assignments in a Windows environment.

Lab Setup

Component	Details
SIEM / Log Analysis	Elastic Stack (Kibana) – `192.168.56.101:5601`
Log Shipper	Elastic Agent (Filebeat) on Windows endpoint
Endpoint	Window-Endpoint (Windows 10)
Audit Policy	Special Logon auditing enabled via `auditpol.exe`

Hypothesis

"An adversary may be abusing valid accounts by assigning special privileges to escalate access or maintain persistence."

- **MITRE ATT&CK Mapping:** T1078 (Valid Accounts)
- **Event Targeted:** Windows Event ID 4672 – Special Privileges Assigned to New Logon

Methodology (SqRR Framework)

Search

- Opened Kibana → Discover
- Selected the `window-security` data view
- Time range: April 28, 2026 (full day)

Query

Executed the following KQL query:

text
```
agent.name: "Window-Endpoint" AND event.code: "4672"
```

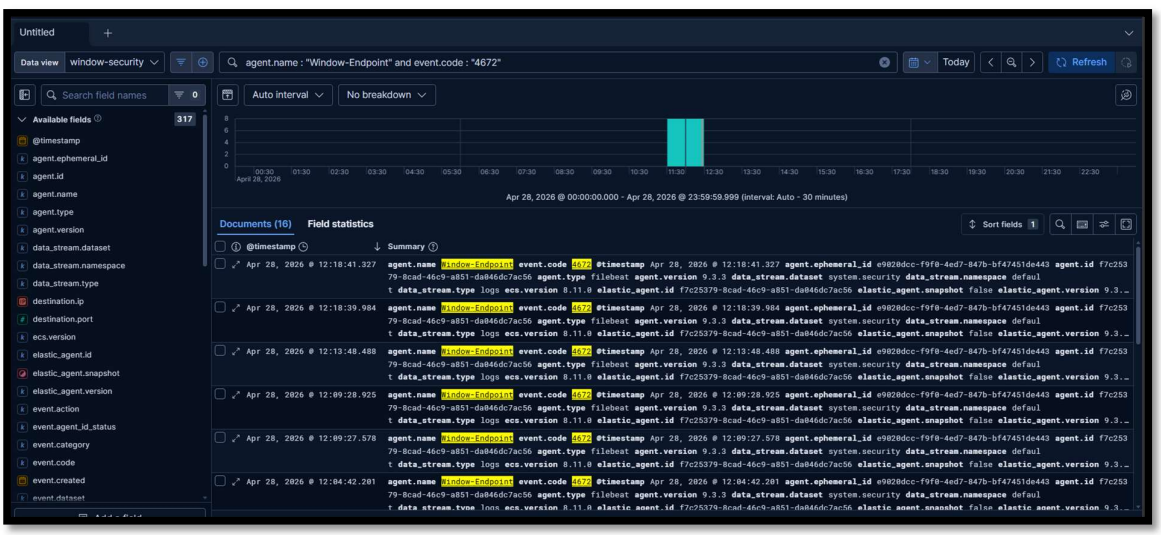
Retrieve

Query returned **16 matching documents** spanning from early morning to afternoon of April 28, 2026.

Representative findings:

Timestamp	Agent	Event Code	Dataset	Notes
2026-04-28 12:09:28	Window-Endpoint	4672	system.security	Special logon detected
2026-04-28 12:13:48	Window-Endpoint	4672	system.security	Special logon detected
2026-04-28 12:18:39	Window-Endpoint	4672	system.security	Special logon detected
2026-04-28 12:18:41	Window-Endpoint	4672	system.security	Special logon detected
2026-04-28 12:29:27	Window-Endpoint	4672	system.security	Special logon detected
2026-04-28 12:29:28	Window-Endpoint	4672	system.security	Special logon detected
2026-04-28 12:44:42	Window-Endpoint	4672	system.security	Special logon detected

Screenshot:



Respond

All observed events must be correlated with authorized administrative activities. Any privilege assignment not matching an approved change window or administrative task should trigger an incident response process, including account verification and session termination.

Observations

- 16 events observed over the day

- All originated from `Window-Endpoint`
- Normal pattern for an active Windows system with regular administrative tasks (Run as Administrator, scheduled tasks, services)
- In a real SOC environment, each event's `SubjectUserName` and `PrivilegeList` fields would be examined for anomaly detection

Conclusion

Threat hunting for T1078 using Windows Event 4672 is an effective baseline technique. The query successfully returned results, demonstrating proper log collection, parsing, and SIEM integration. The hypothesis is validated — this method can identify privileged session creation. Continuous monitoring and correlation with PAM (Privileged Access Management) logs would enhance detection accuracy.

Lessons Learned

- Audit policy must be explicitly enabled for Event 4672 to appear
- Kibana Discover provides rapid ad-hoc hunting capabilities
- Combining MITRE ATT&CK with specific Event IDs creates focused, efficient hunts